Lewis Structures (Kelter) — step 0 for (1–8)

1. Sum valence e- (**= group #**; add 1 per −, remove 1 per + for ions)
2. Sum "wants" (table; period 3+ central: see expanded octet below)
3. Bonds = (wants − valence) ÷ 2 (round down)
4. Central atom = lowest EN (**never H**); ions go in brackets, charge outside
5. Remaining e- = valence − 2×bonds: outer atoms first, then central; double/triple bond if central lacks octet or nonzero FC
6. FC = valence − (lone e- + # bonds): want overall 0, else − on most EN atom

	H	Be	B, Al	C, Si	N, P	O, S	F–I	Noble
Val e-	1	2	3	4	5	6	7	8
Wants	2	4	6	8	8	8	8	8

- **Expanded octet** — period 3+ central (P, S, Cl, Br, I, Xe) can hold >8, so step 3 can undercount bonds. If bonds < # outer atoms: ignore step 3 — single-bond **every** outer atom, complete outer octets, then put **all leftover e- on the central** as lone pairs
- ClF3: 28 e- − 3 bonds (6) − 9 F LP (18) = 4 left → **2 LP on Cl** → 5 groups → T-shaped. Likewise SF4 (1 LP, seesaw); XeF4 (2 LP, sq. planar); XeF2, I3- (3 LP, linear); PCl5, SF6 (0 LP)

Periodic Trends

	Trend	Top	Bot	Left	Right
Atomic size (radius)	↙ (Cs)	↓	↑	↑	↓
EN (electronegativity)	↗ (F)	↑	↓	↓	↑
IE (ionization energy)	↗ (He)	↑	↓	↓	↑
EA (electron affinity)	↗ (Cl)	↑	↓	↓	↑

- Cation < neutral < anion; ↑ΔEN → ↑ bond dipole (3–4); smaller ion, higher charge → ↑ΔHhydr (18); larger atom → ↑ dispersion (9–12)

Trend	Bond	Enthalpy	Exception
EA (X + e- → X-)	form	**exothermic**	endo: Be, Mg, noble, EA2 (usually)
IE	break	**endothermic**	—

- **Lattice energy** U ∝ Q1·Q2 / (r1 + r2): heat **released** (exo.) forming ionic solid from gaseous ions (*dissociation* = endo); max U = ↑ charges, ↓ radii; used in ΔHsoln (18)

(1–2) Molecular Geometry (VSEPR)

- **Central atom** BG + LP (double/triple bond or radical e- = 1 group); lone pairs compress angles (SCl2 a little < 109.5°); compound examples in the (3–4) table

EG	e- geo.	Hyb.	BG	LP	Mol. geo.	Angle	Polarity
2	linear	sp	2	0	linear	180°	nonpolar
3	tri. planar	sp2	3	0	tri. planar	120°	nonpolar
3	tri. planar	sp2	2	1	bent	<120°	polar
4	tetra.	sp3	4	0	tetra.	109.5°	nonpolar
4	tetra.	sp3	3	1	tri. pyr.	<109.5°	polar
4	tetra.	sp3	2	2	bent	<109.5°	polar
5	tri. bipyr.	sp3d	5	0	tri. bipyr.	90/120°	nonpolar
5	tri. bipyr.	sp3d	4	1	seesaw	90/120°	polar
5	tri. bipyr.	sp3d	3	2	T-shaped	90, 180°	polar
5	tri. bipyr.	sp3d	2	3	linear	180°	nonpolar
6	octa.	sp3d2	6	0	octa.	90°	nonpolar
6	octa.	sp3d2	5	1	sq. pyr.	90°	polar
6	octa.	sp3d2	4	2	sq. planar	90°	nonpolar

(3–4) Polarity & Dipole Moment

Compounds	BG	LP	Shape	Polarity
diatomic: H2, Br2 (same); CO, IBr, HCl (diff.)	1	—	linear	nonpol. / pol.
BeCl2, CO2, CS2	2	0	linear	nonpolar
BF3, BH3, SO3	3	0	tri. planar	nonpolar
SO2, O3, ClNO	2	1	bent	polar
CH4, CCl4, CF4, SiH4, PO43-, hydrocarb.	4	0	tetra.	nonpolar
CH3Cl/F/Br/I, CH2Cl2, CHCl3 (mixed)	4	0	tetra.	polar
NH3, PH3, AsH3, PCl3, NCl3, NF3	3	1	tri. pyr.	polar
H2O, OF2, H2S, SCl2, NH2-	2	2	bent	polar
PCl5, PBr5	5	0	tri. bipyr.	nonpolar
ClF3	3	2	T-shaped	polar
XeF2, I3-	2	3	linear	nonpolar
BrF5	5	1	sq. pyr.	polar
XeF4	4	2	sq. planar	nonpolar

- **Dipole** = separation of + / − charge
- (μ ≠ 0) = **polar** = bond dipoles (ΔEN) **don't cancel** (lone pairs or mixed outer atoms)
- CH3**F** > CH3Cl > CH3Br > CH3I; NH3 > PH3 > AsH3 (EN)
- Polar dissolves polar: NH3 in H2O ✓, H2S in NH3 ✓; CCl4, SiO2, O2, CO2 ✗
- Solvent polarity: H2O > ethanol > acetone > CH2Cl2 > diethyl ether > toluene > hexane
- **Nonpolar** Alkanes CH3 [CH2(n-1)] [CH3/OH] (meth 1 (CH4), eth 2, prop 3, but 4, pent 5, hex 6, hept 7, oct 8, non 9, dec 10)

(5–6) Hybridization of an Atom in a Structure

- Look up the EG **of that atom** in the (1–2) table: 2 → sp, 3 → sp2, etc.
- C, 4 single bonds → **sp3**; C with 1 double bond (C=O, C=C, benzene ring) → **sp2**; C with triple or 2 doubles → **sp**
- O in –O–H or ether (2 atoms + 2 LP) → **sp3**; carbonyl =O → **sp2**
- N with 3 atoms + 1 LP (amine) → **sp3**
- Each atom separately: CH3–CHO → left C sp3, carbonyl C sp2

(7–8) σ and π Bonds: Counting & Concepts

- **single = 1σ; double = 1σ + 1π; triple = 1σ + 2π**
- **σ** = hybrid orbitals, head-on along the bond axis; **π** = unhybridized p-p, above & below the axis
- **cis** = same side, **trans** = opposite sides; **π restricts rotation** (double/triple bonds)
- CH3–CH=CH–COOH: 11σ, 2π; benzene C6H6: 12σ (6C–C + 6C–H) + 3π
- Two 1s (or head-on p+p along bond axis) → σ and σ*; sideways p+p (perpendicular axis) → π and π*
- Benzene delocalized π MOs: e- **free to move around the ring**

(9–10) Intermolecular Forces

Force	Found in
Dispersion (London, van der Waals, induced dipole)	**ALL** molecules; ↑ w/ size & molar mass; straight-chain > branched
Dipole-dipole	polar molecules
H-bond	H bonded **directly to N, O, F**
Ion-dipole	ion + polar solvent (NaCl in H2O)

Compounds	Disp.	Dip.-dip.	H-bond
Br2, Cl2, F2, H2, CO2, BCl3, CF4, hydrocarb., noble gases	✓	✗	✗
HCl, HBr, SO2, CHCl3, CH3Br, CH3CHO, CH3CN, ethers (CH3OCH3), H2S, CH3SCH3	✓	✓	✗
H2O, NH3, HF, CH3OH, CH3CH2OH, CH3NH2, CH3COOH, H2O2, HOCH2CH2OH	✓	✓	✓

(11–12) Rank Vapor Pressure & Boiling Point

- **Ionic > H-bond > dipole-dipole > dispersion** (at similar size)

IMF	bp	mp	Visc.	Surf. tens.	ΔHvap	Crit. T	VP
↑	↑	↑	↑	↑	↑	↑	↓
↓	↓	↓	↓	↓	↓	↓	↑

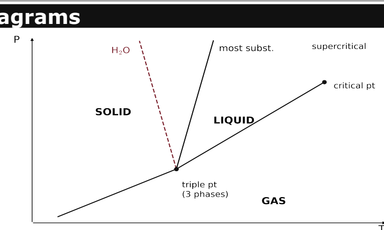
Rank	Example
VP ↓	CF4 > PF5 > BrF3 (polar = lowest VP)
bp ↑	CH4 < CH3Cl < CH3OH < RbCl (ionic)
bp (disp. size)	I2 > Br2 > Cl2 > F2; CH3CHO > CH3OCH3 > Rn
bp (H-bond)	HF > HI > HBr > HCl
bp (# of OH)	HOCH2CH2OH > ClCH2CH2OH > CH3CH2OH > CH3CH2Cl

(13) Viscosity, Capillary Action, Surface Tension

Surface tension	energy required to stretch/increase surface **area** by unit amount
Viscosity	resistance of a liquid to **flow**; ↑ with IMF, ↓ with ↑T; spherical/compact → lower
Capillary action	liquid flows **up a thin tube against gravity** = cohesion (**like** molecules) + adhesion (liquid–glass)

- Concave meniscus (H2O): adhesion > cohesion; convex (Hg): cohesion > adhesion. Highest viscosity = most H-bonding (HOCH2CH2OH)

(14) Phase Diagrams



- Lines = 2 phases coexist; **triple point** = all 3; **critical point**: above Tc no pressure can liquefy gas → **supercritical fluid**
- ↑T at const P (→): solid → liquid → gas. ↑P at const T (↑): gas → liquid → solid. Normal bp/mp read at 1 atm

Endothermic (+ΔH, +ΔS)	Exothermic (−ΔH, −ΔS)
melting s→l	freezing l→s
vaporization l→g	condensation g→l
sublimation s→g	deposition g→s

- Order: solid = greatest, gas = least (↑T = ↓ order)
- **H2O**: fusion line leans **left** (ice less dense) → ↑P melts ice. **CO2**: triple pt at 5.1 atm → at 1 atm solid **sublimes** s→g (dry ice)

(15–16) Type of Solid

Type	Held by	Properties	Examples
Ionic	± attraction	hard, brittle, high mp, poor cond.	NaCl, AgCl, CaF2, MgO, CaCO3
Molecular	IMFs	soft, low mp, poor cond.	ice H2O, CO2, I2, sucrose
Metallic (atomic)	e- sea	low–high mp, **good cond.**	Au, Cu, Fe, Na
Network cov. (atomic)	covalent bonds	v. hard, v. high mp, poor cond.	diamond, quartz SiO2, graphite, SiC
Nonbonding (atomic)	dispersion	v. low mp	noble gases: Ar(s)
Amorphous	no long-range order	melts over a **range**	glass, rubber, plastic, butter, wax
Ceramic	inorganic nonmetal, fired	brittle, heat-resistant	clay/kaolinite, porcelain, Al2O3

- Graphite sp2 sheets: conducts along, slippery; diamond sp3: insulator
- FCC packing = most space-efficient (12 neighbors)

(17) Band Theory & Doping

- Bands = MOs delocalized over the whole crystal: **valence band = bonding MOs**, **conduction band = antibonding MOs**, **band gap** between
- **Conductor** (metal): bands continuous/overlap; **Semiconductor** (metalloids Si, Ge): small gap, e- promotable; **Insulator: LARGE gap**
- **Doping** = add impurity to ↑ conductivity:

	Dopant	Carrier
n-type	Group **V/5A** (P, As)	extra e- in **conduction** band (− charges)
p-type	Group **III/3A** (B, Al, Ga, In)	**holes** in **valence** band (e- hop between holes)
p–n junction	—	**diode**: current flows one way

(18) Heat of Solution & Hydration

$$\Delta H_{solution} = \Delta H_{hydration} - \Delta H_{lattice\ energy}$$

$$\Delta H_{solution} = \Delta H_{solute} + (\Delta H_{solvent} + \Delta H_{mix})$$
$$= (-\Delta H_{lattice}) + \Delta H_{hydration}$$

- Dissolving = break crystal (**ΔHsolute = −ΔHlattice**, costs energy) + make ion-dipole attractions (**ΔHsolvent + ΔHmix = ΔHhydration**, releases)
- **ΔHhydration**: heat **released** when 1 mol **gaseous ions** dissolves in water (always exo)
- **Highest ΔHhydr**: smallest ion + highest charge: Al3+ > Mg2+ > Na+ > K+; F- > Cl-
- ΔHsoln **exo (−)**: ΔHhydration > ΔHsolute; **endo (+)**: solute > hydration; **≈ 0**: comparable

(19) Saturation; Like Dissolves Like

Unsaturated	less than equilibrium amount — more can dissolve
Saturated	max amount at that T; dissolved ⇌ solid **dynamic equilibrium**
Supersaturated	**more** than equilibrium amount; unstable — seed crystal/ scratch → excess crystallizes out

- Solution forms if solute-solvent attraction **comparable** to solute-solute + solvent-solvent; polar/ionic ↔ polar (H2O, ethanol, NH3); nonpolar ↔ nonpolar (hexane, CCl4, benzene)
- Most soluble in ethanol = most H-bonds (ethylene glycol); in hexane = longest chain (1-pentanol); all **nitrates** soluble

(20) Solubility vs P & T; When Bubbles/Boiling

↑ gives	Gas	Solid / liquid
↑ **Pressure**	↑ solubility (Henry)	no effect
↑ **Temperature**	↓ solubility	↑ solubility

- Dissolve the most gas: **low T + high P** (cold soda holds CO2)
- Liquid **boils** when its vapor pressure **= external (atm) pressure**; gas **bubbles out** of solution when its partial (vapor) pressure **> atm pressure**

(21) Molarity (and Molality)

$$M = \frac{mol\ solute}{L\ solution} \qquad m = \frac{mol\ solute}{kg\ solvent}$$

- mL ÷ 1000 = L; g ↔ mol via molar mass; dilution: M1V1 = M2V2
- mol = M × V(L): 0.500 M × 0.250 L = **0.125 mol**
- V = mol ÷ M: 0.300 mol ÷ 0.450 M = 0.667 L = **667 mL**
- Molality m uses **kg of solvent**, not solution — for ΔTf/ΔTb
- **Only molarity changes with temperature** (volume swells); molality, mass %, χ don't (mass/count-based)

(22) Percent Concentration (% w/v)

$$\%\,w/v = \frac{g\ solute}{100\ mL\ soln} \ \Rightarrow\ g = \frac{\% \times mL}{100}$$

- g = % × mL ÷ 100: for 250 mL of 1.5% → 1.5 × 250 ÷ 100 = **3.75 g** (the summary's "31.25 g" contradicts its own formula)
- mL = g ÷ (% ÷ 100): 16.6 g ÷ 0.0450 g/mL = **369 mL**
- **% m/m** = g solute ÷ (g solute + g solvent) × 100 — denominator = **whole solution**
- g water = g solution × (1 − % ÷ 100): 300 g of 9.71% KBr → 300 × 0.9029 = **271 g water**
- **% v/v** = vol component ÷ total vol × 100

(23) Henry's Law (Cgas = kH·Pgas)

$$C_{gas} = k_H\,P_{gas} \qquad k_H = C_{gas}/P_{gas}$$

- kH = "solubility constant of a gas" (M/atm); C = molar solubility
- **kH = C ÷ P** (divide — below 1 atm, kH comes out **larger** than C)
- **1 atm** = 760 mmHg = 760 torr = 14.7 psi = 101.3 kPa (mmHg → atm: **÷ 760**)

Given C = 4.7e−4 mol/L at 522 mmHg:

1. P = 522 ÷ 760 = 0.687 atm
2. kH = 4.7e−4 ÷ 0.687 = **6.8e−4 mol/L·atm**

(24) Osmosis

- **Osmosis**: **solvent** (water) flows across a semipermeable membrane (passes solvent, **blocks solute** ions — it does *not* stop water) from **low solute conc. → high solute conc.**
- **Hyperosmotic** = more solute: water flows **into** it (cell shrinks)
- **Hyposmotic** = less solute: water flows **out** to the more concentrated side (cell swells)
- **Isosmotic** = equal: no net flow (isotonic saline = 0.9% w/v NaCl); π = pressure needed to **stop** osmotic flow

(25) MO Theory — Bond Order & Magnetism

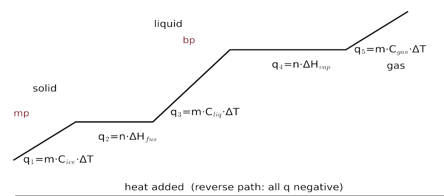
	B2, C2, N2 pattern	O2, F2, Ne2 pattern
high	σ*2p	σ*2p
↑	π*2p, π*2p	π*2p, π*2p
↑	**σ2p**	**π2p, π2p**
↑	**π2p, π2p**	**σ2p**
↑	σ*2s	σ*2s
low	σ2s	σ2s
Used by	B2, C2, N2, N2+, N2-	O2, O2+, O2-, F2, Ne2; **NO, NO+, NO-** (heteronuclear: use the more EN atom's pattern); Cl2, Cl2+, Cl2- (period 3: n = 3)

- 1s 2s 2p 3s 3p 4s 3d 4p 5s 4d 5p 6s 4f 5d 6p 7s 5f 6d 7p

1. Valence e- total: **add** 1 per − charge, **remove** 1 per +
2. Fill round-robin sublevel before double occupation (Hund)
3. **BO = (bonding e- − antibonding e-) / 2**
4. Any unpaired e- → **paramagnetic**; all paired → **diamagnetic**
5. BO > 0 → stable, can form; **higher BO = stronger & shorter** bond (N2 > N2+). Counting 2p alone gives same BO (σ2s + σ*2s cancel)
6. **Heteronuclear** (NO, CO, OF-): sum both atoms' valence e-, fill the **O2 pattern**. NO 11 e- → BO 2.5, para; NO+ 10 → 3, dia; NO- 12 → 2, para
7. Two-sided diagram: **more EN atom's orbitals sit lower** (more stable); bonding MOs resemble it

Sp.	e-	BO	Mag	Sp.	e-	BO	Mag
N2	10	3	dia	O2	12	2	**para** (2↑)
N2+	9	2.5	para	O2+	11	2.5	para
N2-	11	2.5	para	O2-	13	1.5	para
F2, Cl2	14	1	dia	Cl2+	13	1.5	para
B2	6	1	para	Cl2-	15	0.5	para

(26) Phase-Change Heat (gas ⇄ solid)



1. Sketch the path; skip segments you don't cross (see examples below)
2. Sloped (one phase): **q = m(g)·Cs·ΔT**; flat (phase change): **q = n(mol)·ΔH**
3. Sum; convert J → kJ. Cooling path (gas → solid): every q **negative** (ΔHcond = −ΔHvap, ΔHfreeze = −ΔHfus)

- H2O: C(ice) 2.09, C(liq) 4.184, C(steam) ≈ 2.0 J/g·°C; ΔHfus 6.02, ΔHvap 40.7 kJ/mol; MM 18.02

36.0 g H2O, liquid 65 °C → gas 115 °C:

1. heat liquid 65→100: q = 36.0 g × 4.184 × 35° = 5.27 kJ
2. boil (use mol): q = (36.0 ÷ 18.02) × 40.7 = 81.3 kJ
3. heat gas 100→115: q = 36.0 g × 2.01 × 15° = 1.09 kJ
4. total = **87.7 kJ**

- Cooling runs the same steps backward: 125 g benzene (MM 78.11), gas 425 K → liquid 335 K (bp 353, mp 279 — it never freezes): 9.5 kJ + 54.3 kJ + 3.9 kJ = **67.7 kJ removed**
- ΔHsub = ΔHfus + ΔHvap (endo/exo table in (14))

(27) Raoult's Law (VP Lowering)

$$P_{soln} = \chi_{solvent}\,P^{\circ}_{solvent} \qquad \Delta P = \chi_{solute}\,P^{\circ}_{solvent}$$

$$\chi_{solvent} = \frac{mol_{solvent}}{mol_{solvent} + i \cdot mol_{solute}}$$

1. mol solvent = g ÷ MM; mol solute = g ÷ MM (**× i if ionic**)
2. χ(solvent) = mol solvent ÷ total mol — always < 1
3. P(soln) = χ(solvent) × P°; lowering ΔP = P° − P(soln) = χ(solute)·P°

- 75.0 g citric acid (192.12) + 420 g H2O, P° 71.93 mmHg: 0.390 & 23.31 mol → χ = 0.9835 → **70.7 mmHg**

(28) Freezing ↓ / Boiling ↑ (van't Hoff)

$$\Delta T_f = i\,K_f\,m \qquad \Delta T_b = i\,K_b\,m$$

- m = mol solute / **kg solvent**; H2O: Kf 1.86, Kb 0.512 °C/m; Tf = T°f **−** ΔTf, Tb = T°b **+** ΔTb
- **i** = particles per formula unit: glucose, sucrose 1; NaCl, KCl, NaOH 2; CaCl2, (NH4)2SO4 3; Al(NO3)3 4. Given the bp: i = ΔTb ÷ (Kb × m)
- Lowest fp for equal grams = small MM and high i (NaOH), **but the HW answer key rewards the most ions per formula (CaCl2, i = 3)**
- Molar mass from fp — 20.0 g unknown per 500 g benzene (T°f 5.444, Kf 5.12) freezes at 3.77 °C:

1. ΔTf = 5.444 − 3.77 = 1.674 °C
2. m = ΔTf ÷ (i × Kf) = 1.674 ÷ 5.12 = 0.327 mol/kg
3. mol = m × kg solvent = 0.327 × 0.500 = 0.163 mol
4. MM = g ÷ mol = 20.0 ÷ 0.163 = **≈120 g/mol**

(29) Osmotic Pressure → Molar Mass

$$\pi = i\,MRT \ \Rightarrow\ M = \frac{\pi}{i\,RT}$$

- R = **0.08206** L·atm/mol·K; T in **K** (+273.15); π in **atm** (torr ÷ 760); i = 1 nonelectrolyte

Given 15.2 **mg** in 150.0 mL, π = 8.44 torr, 25 °C:

1. M = π ÷ (iRT) = (8.44 ÷ 760) ÷ (0.08206 × 298) = 4.54e−4 M
2. mol = M × V(L) = 4.54e−4 × 0.1500 = 6.81e−5 mol
3. MM = g ÷ mol = 0.0152 ÷ 6.81e−5 = **223 g/mol**